A conditional BRST gauge-fixing and Faddeev–Popov closure note for the TECT defect sector, with a TECT-specific Berry correction from Math160

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We present a candidate BRST gauge-fixing and Faddeev–Popov closure note for the GAP-2 quantum-completion gate in the Topological Energy Condensate Theory (TECT) defect sector. The construction combines (a) the standard BRST / Faddeev–Popov path-integral measure for the gauge field in covariant gauge (textbook QFT input) and (b) a candidate TECT-specific Berry-phase correction inherited from Math160 arising from the adiabatic evolution of the BCC order parameter during freeze-out. The current canonical tier of GAP-2 is **T6** per Math307 / Math310, conditional on Pillar 4's stability theorem (Math80-AddA, hypothesis H1.1) via the inheritance mechanism of Math280.

Honest scope: this draft is a conditional closure note, NOT a stand-alone derivation. The novelty content lies in the Berry-correction inheritance from Math160; the present draft cites Math160 rather than re-deriving the correction, and the standard BRST/FP textbook part is sketched at programme level rather than fully written out with Grassmann measure and sign-convention discipline. The over-claim wording ("GAP-2 is now complete", "fully rigorous", "ready for publication") of an earlier draft has been removed; promotion to a stand-alone derivation requires (i) full BRST/FP textbook write-up with Grassmann measure, (ii) explicit re-derivation of the Math160 Berry correction within the present paper, (iii) consistency cross-check with the canonical GAP-2 tier T6 in `Docs/status/TOE-FACT-SHEET.md`.

I. INTRODUCTION

GAP-2 addresses a textbook quantum-field-theoretic issue, the gauge-fixed path-integral measure for non-Abelian gauge theories. The standard treatment proceeds via the Faddeev–Popov determinant [1], which introduces auxiliary anti-commuting ghost fields to compensate for the over-counting introduced by gauge fixing, and via the BRST cohomology [2, 3], which encodes the residual rigid symmetry that selects the physical Hilbert subspace.

In the TECT defect-sector context, the gauge field of Pillar 4 acquires the same Faddeev–Popov / BRST structure as the SM gauge sector; the present paper records the inheritance of that structure

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under the conditional Pillar 4 closure (hypothesis H1.1 of Math80-AddA), supplemented by a candidate TECT-specific Berry-phase correction inherited from Math160 due to the adiabatic evolution of the BCC order parameter through the defect manifold. The present paper does *not* re-derive the textbook BRST/FP machinery; it inherits it under hypothesis (H1) and combines it with the Math160 Berry inheritance under hypothesis (H2). The closure is *conditional*, not stand-alone.

II. SETTING — BRST / FADDEEV–POPOV INHERITANCE IN COVARIANT GAUGE

We work throughout in the Minkowski signature $(+,-,-,-)$ with metric $\eta_{\mu\nu}$ and adopt the standard non-Abelian Yang–Mills action

$$S_{\rm YM}[A] \;=\; -\frac{1}{4}\int d^4x\, F^a_{\mu\nu}\, F^{a\,\mu\nu}, \qquad F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g\, f^{abc} A^b_\mu A^c_\nu, \tag{1}$$

with $\{T^a\}$ the gauge-group generators in the matter representation, f^{abc} the structure constants, and a, b, c running over the adjoint index of the Pillar 4 gauge sector.

The covariant gauge-fixing functional is $G^a(A) = \partial^\mu A^a_\mu$, with parameter ξ. The Faddeev–Popov [1] insertion of the identity

$$1 \;=\; \det\!\Big(\tfrac{\delta G^a(A^\omega)}{\delta\omega^b}\Big) \int \mathcal{D}\omega\;\delta\big(G^a(A^\omega) - \chi^a\big), \tag{2}$$

followed by Gaussian-averaging χ^a with weight $\exp[-\frac{i}{2\xi}\int d^4x(\chi^a)^2]$ and exponentiation of the determinant via auxiliary anti-commuting (Grassmann) ghost fields $c^a, \bar c^a$, gives the standard gauge-fixed gauge generating functional

$$Z[J,\bar\eta,\eta] \;=\; \int \mathcal{D}A\,\mathcal{D}\bar c\,\mathcal{D}c\;\exp\!\Big(i\,S_{\rm eff}[A,\bar c,c] + i\int d^4x\big(J^{a\,\mu}A^a_\mu + \bar\eta^a c^a + \bar c^a \eta^a\big)\Big), \tag{3}$$

with the gauge-fixed effective action

$$S_{\rm eff}[A,\bar c,c] \;=\; S_{\rm YM}[A] \;-\; \frac{1}{2\xi}\int d^4x\,\big(\partial^\mu A^a_\mu\big)^2 \;-\; \int d^4x\,\bar c^a\,\partial^\mu D^{ac}_\mu\, c^c, \tag{4}$$

where $D^{ac}_\mu = \partial_\mu \delta^{ac} + g\, f^{abc} A^b_\mu$ is the adjoint covariant derivative. The ghost integration measure $\mathcal{D}\bar c\,\mathcal{D}c$ is the textbook Berezin (Grassmann) measure, with the convention $\int d\bar c\, dc\, \bar c\, c = -1$ per pair so that the exponentiation (3) reproduces the determinant in (2) with a positive sign.

Theorem 1 (BRST / FP closure of the Pillar 4 gauge sector with TECT-specific Math160 Berry correction; **T6 proved conditional on (H1)+(H2)**). *Under the hypotheses*

(***H1***) *the Pillar 4 gauge-stability theorem of Math80-AddA (hypothesis H1.1, the Grassmannian-stability component of the Math80 closure programme);*

(***H2***) *the Math160 Berry-correction inheritance theorem (the BCC order parameter undergoes adiabatic evolution through the Pillar 4 defect manifold and contributes a geometric phase to the ghost effective action),*

the gauge-fixed effective action of (4) *acquires the TECT-specific Berry contribution and reads*

$$\boxed{S_{\rm eff}^{\rm TECT}[A,\bar c,c] \;=\; S_{\rm YM}[A] \;-\; \frac{1}{2\xi}\int d^4x\,(\partial^\mu A_\mu^a)^2 \;-\; \int d^4x\,\bar c^a\,\partial^\mu D_\mu^{ac}\,c^c \;+\; S_{\rm TECT}^{\rm Berry}[A,\bar c,c],} \tag{5}$$

where $S_{\rm TECT}^{\rm Berry}$ is the Math160-derived Berry-phase correction. Under (H1)+(H2), the action (5) *is invariant under the standard nilpotent BRST transformation*

$$sA_\mu^a = D_\mu^{ac}c^c, \quad sc^a = -\tfrac{1}{2}g\,f^{abc}c^bc^c, \quad s\bar c^a = -\tfrac{1}{\xi}\,\partial^\mu A_\mu^a, \quad s^2 = 0, \tag{6}$$

so that the BRST cohomology of (5) *agrees with that of the textbook gauge-fixed Yang–Mills action; in particular, the physical-state subspace is the BRST-closed-modulo-exact subspace of the total Hilbert space, and the S-matrix on physical states is unitary.*

Conditional tier: *GAP-2 is recorded at* ***T6*** **proved conditional** *per Math307 / Math310, conditional on (H1) Pillar 4 H1.1* and *(H2) the Math160 Berry inheritance.*

Remark 2 (Honest scope of the theorem). Theorem 1 *inherits* the textbook BRST / FP machinery rather than re-deriving it within the present paper, and *cites* the Math160 Berry-correction derivation rather than re-deriving it. The novelty content of the present paper is the inheritance argument under (H1)+(H2), not the textbook-side construction. Promotion of GAP-2 to a stand-alone derivation requires (i) full BRST/FP textbook write-up with explicit Grassmann measure, sign, and Wick-rotation conventions; and (ii) explicit re-derivation of $S_{\rm TECT}^{\rm Berry}$ within the present paper. Neither (i) nor (ii) is performed here; the conditional T6 verdict reflects the inheritance, not a stand-alone derivation.

III. PROOF SKETCH UNDER (H1)+(H2)

a. Step 1 (BRST symmetry setup). The nilpotent BRST transformation in the conventions of (6) is the textbook construction [2, 3]. It satisfies $s^2 = 0$ on every field (this is the Jacobi identity for f^{abc} together with the Grassmann anti-commutativity of c^a).

b. Step 2 (Faddeev–Popov implementation). Insertion of the identity (2) into the unrestricted gauge-field path integral, Gaussian-averaging the gauge-fixing functional, and exponentiating the determinant via the Berezin (Grassmann) measure for $c^a, \bar{c}^a$ yields the gauge-fixed effective action (4).

c. Step 3 (Berry-phase inheritance from Math160). Math160 derives the TECT-specific Berry-phase contribution $S_{\text{TECT}}^{\text{Berry}}[A, \bar{c}, c]$. The contribution arises because the BCC order parameter undergoes adiabatic evolution through the Pillar 4 defect manifold during phase-transition freeze-out and acquires a geometric phase encoded in the ghost boundary conditions on the defect manifold. Under hypothesis (H2) the resulting contribution is added to (4) producing (5).

d. Step 4 (BRST closure verification). Under hypothesis (H1), $S_{\text{TECT}}^{\text{Berry}}$ is s-closed (a property derived in Math160 from the gauge-equivariance of the underlying Berry connection). Therefore the total action (5) is BRST-invariant, $s\, S_{\text{eff}}^{\text{TECT}} = 0$, and the BRST cohomology of (5) agrees with that of the textbook gauge-fixed Yang–Mills action.

e. Step 5 (Conditional inheritance from Pillar 4 via Math280). Math280 establishes the inheritance mechanism: under (H1) Pillar 4 hypothesis H1.1, the BRST/FP closure of (5) inherits the Pillar 4 tier. With Pillar 4 at the present canonical T6 PROVED CONDITIONAL, the GAP-2 closure recorded here is **T6 proved conditional on (H1)+(H2)**. No additional hypothesis beyond (H1)+(H2) is required.

IV. NUMERICAL ESTIMATE OF THE BERRY CORRECTION

In the covariant gauge with fixing parameter $\xi = 1$ (Feynman gauge) and at the linearised level $g \to 0$, the ghost piece of (5) reduces to

$$\mathcal{L}_{\text{ghost}}^{(\xi=1,\, g=0)} \;=\; -\,\bar{c}^a\, \partial^\mu \partial_\mu\, c^a \;+\; \delta\mathcal{L}_{\text{TECT}}^{\text{Berry}}, \tag{7}$$

where $\delta\mathcal{L}_{\text{TECT}}^{\text{Berry}}$ is the linearised limit of the Math160 Berry contribution. Numerical evaluation in Math160 gives a relative correction of order $\sim 10^{-2}$ to the ghost propagator at the one-loop matching scale; this is consistent with the $\mathcal{O}(a_{\text{BCC}}^2)$ expectation for a continuum-limit-vanishing geometric correction. The standard Faddeev–Popov ghost propagator is recovered to leading order in a_{BCC}.

V. DISCUSSION — CONDITIONAL TIER, NOT STAND-ALONE DERIVATION

a. What the present note records. Under hypotheses (H1)+(H2), the gauge-fixed action of the Pillar 4 gauge sector takes the form (5) and admits a nilpotent BRST symmetry (6), so the BRST/FP cohomology agrees with that of the textbook gauge-fixed Yang–Mills sector. The GAP-2 canonical tier per Math307 / Math310 is **T6 proved conditional** on (H1) Pillar 4 H1.1 *and* (H2) the Math160 Berry-correction inheritance. Neither hypothesis is discharged in the present paper.

b. What the present note does NOT establish.

1. It does *not* establish a stand-alone BRST/FP derivation of GAP-2: the textbook part is inherited, the Math160 Berry part is inherited (Remark 2).

2. It does *not* discharge Pillar 4 hypothesis H1.1: that is the canonical Pillar 4 sub-task tracked separately in Math80-AddA.

3. It does *not* certify the asymptotic behaviour of $S_{\mathrm{TECT}}^{\mathrm{Berry}}$ in the continuum limit (lattice spacing $\to 0$); the Math160 Berry correction is expected to vanish in that limit at order $\mathcal{O}(a_{\mathrm{BCC}}^2)$, but a continuum-limit constant-bound theorem on this is a separate item.

c. Promotion path to stand-alone derivation. A stand-alone derivation of GAP-2 within a future revision of this paper requires (i) explicit textbook write-up of the BRST/FP machinery including the Grassmann measure, the Euclidean–Minkowski Wick rotation, and the $i\epsilon$ prescription; (ii) explicit re-derivation of $S_{\mathrm{TECT}}^{\mathrm{Berry}}$ within the present paper; (iii) cross-check with the canonical GAP-2 tier T6 in `Docs/status/TOE-FACT-SHEET.md` after promotion. Until those items are completed, the present note remains a *conditional closure note*, not a stand-alone derivation.

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